ID: 67b59a67

Plants like potatoes, tomatoes, and soybeans are susceptible to bacterial wilt disease caused by the bacteria *Ralstonia solanacearum*. A multinational team of scientists led by Zhong Wei studied whether other microbes in the soil might influence the degree to which plants are affected by the disease. The team sampled soil surrounding individual tomato plants over time and compared the results of plants that became diseased with those that remained healthy. They concluded that the presence of certain microbes in the soil might explain the difference between healthy and diseased plants.

Which finding, if true, would most directly support the team's conclusion?

A. The soil surrounding healthy plants contained significantly higher concentrations of microbes known to inhibit *Ralstonia solanacearum* than the soil surrounding diseased plants did.

B. The soil surrounding the plants contained high concentrations of *Ralstonia solanacearum* regardless of whether the plants were affected by wilt disease.

C. The soil surrounding healthy plants tended to have significantly higher moisture levels than the soil surrounding diseased plants did.

D. By the end of the experiment, over half the plants had been affected by wilt disease regardless of differences in the types and concentrations of microbes in the surrounding soil.

ID: e677fa6c

The following text is adapted from Edgar Allan Poe's 1849 story "Landor's Cottage."

> During a pedestrian trip last summer, through one or two of the river counties of New York, I found myself, as the day declined, somewhat embarrassed about the road I was pursuing. The land undulated very remarkably; and my path, for the last hour, had wound about and about so confusedly, in its effort to keep in the valleys, that I no longer knew in what direction lay the sweet village of B——, where I had determined to stop for the night.

Which choice best states the main idea of the text?

A. The narrator remembers a trip he took and admits to getting lost.

B. The narrator recalls fond memories of a journey that he took through some beautiful river counties.

C. The narrator describes what he saw during a long trip through a frequently visited location.

D. The narrator explains the difficulties he encountered on a trip and how he overcame them.

ID: 95146ebb

The ice melted on a Norwegian mountain during a particularly warm summer in 2019, revealing a 1,700-year-old sandal to a mountaineer looking for artifacts. The sandal would normally have degraded quickly, but it was instead well preserved for centuries by the surrounding ice. According to archaeologist Espen Finstad and his team, the sandal, like those worn by imperial Romans, wouldn't have offered any protection from the cold in the mountains, so some kind of insulation, like fabric or animal skin, would have needed to be worn on the feet with the sandal.

What does the text indicate about the discovery of the sandal?

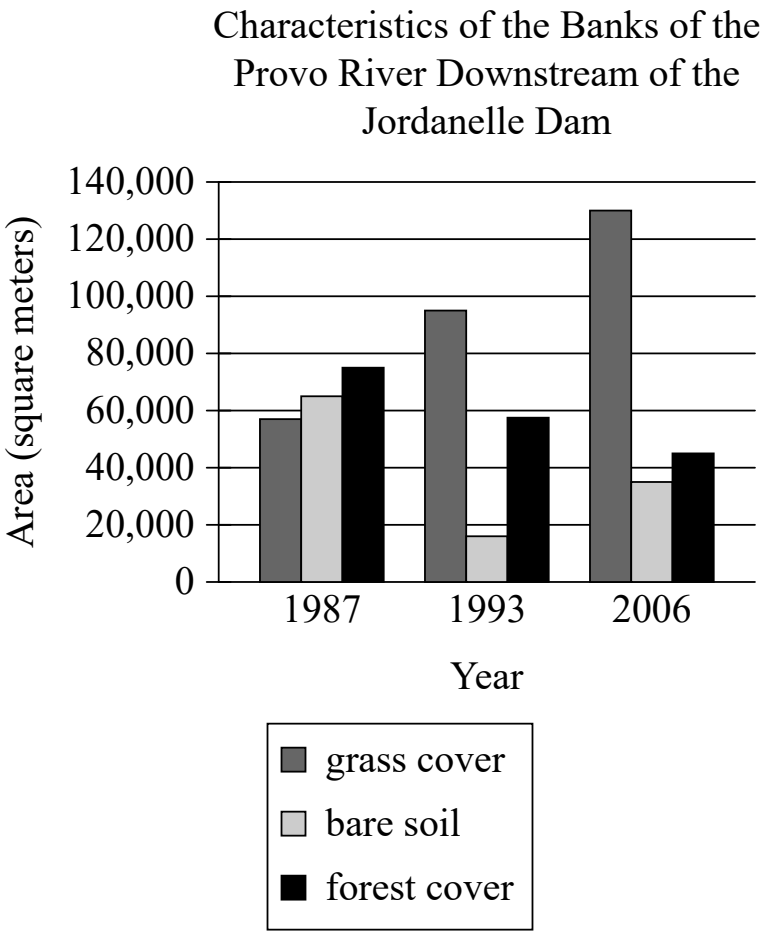
A. Temperatures contributed to both protecting and revealing the sandal.

B. The discovery revealed that the Roman Empire had more influence on Norway than archaeologists previously assumed.

C. Archaeologists would have found the sandal eventually without help from the general public.

D. The sandal would have degraded if it hadn't been removed from the ice.

ID: d83c3d54



The Jordanelle Dam was built on the Provo River in Utah in 1992. Earth scientist Adriana E. Martinez and colleagues tracked changes to the environment on the banks of the river downstream of the dam, including how much grass and forest cover were present. They concluded that the dam changed the flow of the river in ways that benefited grass plants but didn't benefit trees.

Which choice best describes data from the graph that support Martinez and colleagues' conclusion?

A. The lowest amount of grass cover was approximately 58,000 square meters, and the highest amount of forest cover was approximately 75,000 square meters.

B. There was more grass cover than forest cover in 1987, and this difference increased dramatically in 1993 and again in 2006.

C. There was less grass cover than bare soil in 1987 but more grass cover than bare soil in 1993 and 2006, whereas there was more forest cover than bare soil in all three years.

D. Grass cover increased from 1987 to 1993 and from 1993 to 2006, whereas forest cover decreased in those periods.

ID: adbcbce0

The following text is adapted from Christina Rossetti's 1881 poem "Monna Innominata 2."

I wish I could remember that first day,
  First hour, first moment of your meeting me,
  If bright or dim the season, it might be
Summer or Winter for [all] I can say;
So unrecorded did it slip away,
  So blind was I to see and to foresee,
  So dull to mark the budding of my tree
That would not blossom yet for many a May.

Which choice best states the main idea of the text?

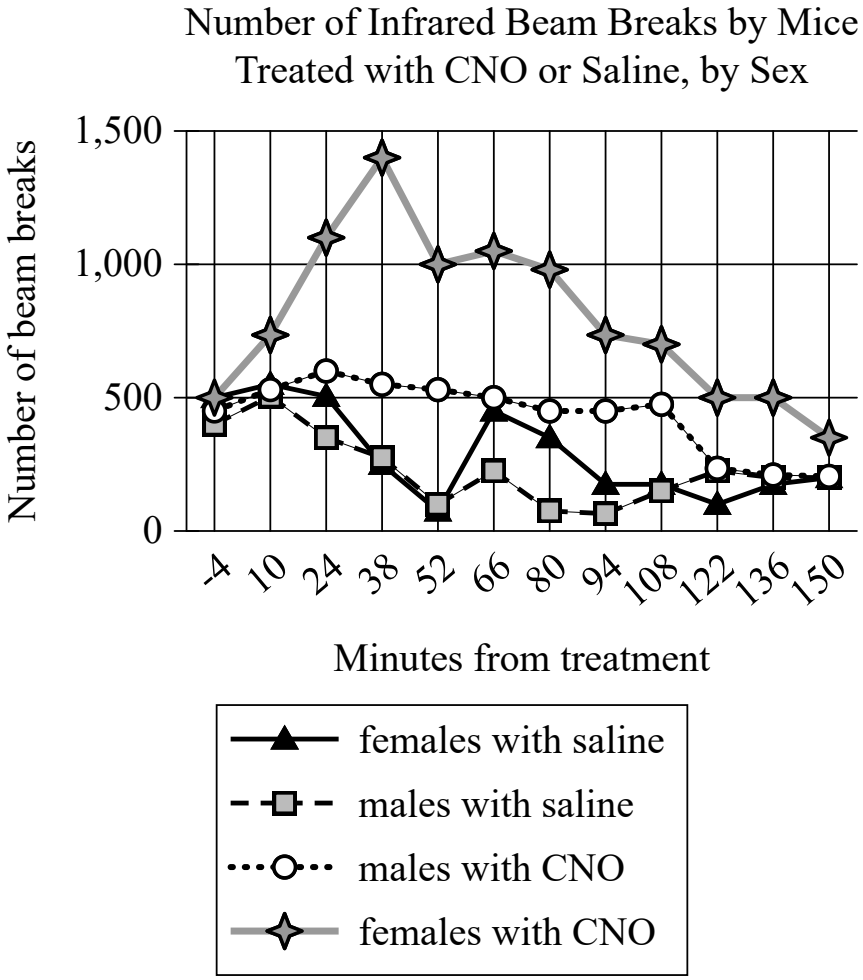
A. The speaker celebrates how the passage of time has strengthened a relationship that once seemed unimportant.

B. Because the speaker did not anticipate how important a relationship would become, she cannot recall how the relationship began, which she regrets.

C. As the anniversary of the beginning of an important relationship approaches, the speaker feels conflicted about how best to commemorate it.

D. After years of neglecting a once valuable relationship, the speaker worries it may be too late for her to salvage the relationship.

ID: e2829dd7



To investigate the influence of certain estrogen-responsive neurons on energy expenditure, biologist Stephanie Correa et al. treated female and male mice with either saline solution or clozapine-N4-oxide (CNO), which activates the neurons. Monitoring the activity levels of the mice by measuring how frequently the animals broke infrared beams crossing their enclosures, Correa et al. found that the mice in their study showed sex-specific differences in response to neuron activation: _____

Which choice most effectively uses data from the graph to complete the assertion?

A. the four groups of mice differed greatly in their activity levels before treatment but showed identical activity levels at the end of the monitoring period.

B. saline-treated females showed substantially more activity at certain points in the monitoring period than saline-treated males did.

C. CNO-treated females showed more activity relative to saline-treated females than CNO-treated males showed relative to saline-treated males.

D. CNO-treated females showed a substantial increase and then decline in activity over the monitoring period, whereas CNO-treated males showed a substantial decline in activity followed by a steep increase.

ID: dc47c2ac

Although most songbirds build open, cupped nests, some species build domed nests with roofs that provide much more protection. Many ecologists have assumed that domed nests would provide protection from weather conditions and thus would allow species that build them to have larger geographic ranges than species that build open nests do. To evaluate this assumption, a research team led by evolutionary biologist Iliana Medina analyzed data for over 3,000 species of songbirds.

Which finding from Medina and her colleagues' study, if true, would most directly challenge the assumption in the underlined sentence?

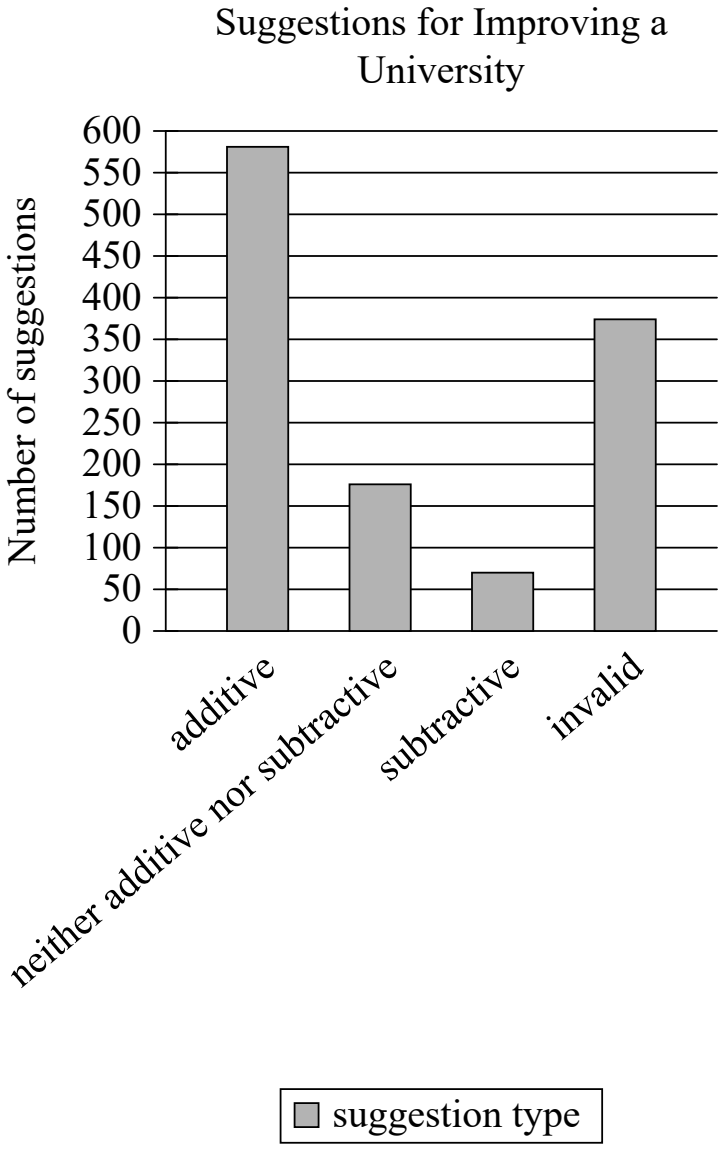
A. Species that build open nests tend to have higher extinction rates than species that build domed nests.

B. Species that build open nests tend to be smaller in size than species that build domed nests.

C. Species that build open nests tend to use fewer materials to build their nests than species that build domed nests do.

D. Species that build open nests tend to have larger ranges than species that build domed nests.

ID: 23e2421a



Gabrielle Adams and colleagues reviewed suggestions for improving a university that had been submitted to the university's president. They coded each suggestion as additive (the idea suggested adding something new to the university), subtractive (the idea suggested removing something from the university), neither additive nor subtractive, or invalid (the idea was not comprehensible). The data illustrated people's tendency to overlook the possibility of removing things to achieve improvements: _____

Which choice most effectively uses data in the graph to complete the statement?

A. around 175 suggestions were coded as neither additive nor subtractive, whereas around 575 suggestions were coded as additive.

B. more than 350 suggestions were coded as invalid, whereas fewer than 100 suggestions were coded as subtractive.

C. fewer than 100 suggestions were coded as subtractive, whereas more than 550 suggestions were coded as additive.

D. around 575 suggestions were coded as additive, whereas around 175 suggestions were coded as subtractive.

ID: ad680167

The recovery of a 1,000-year-old Chinese shipwreck in the Java Sea near present-day Indonesia has yielded a treasure trove of artifacts, including thousands of small ceramic bowls. Using a portable X-ray fluorescence analyzer tool, Lisa Niziolek and her team were able to detect the chemical composition of these bowls without damaging them. By comparing the chemical signatures of the bowls with those of the materials still at old Chinese kiln sites, Niziolek and her team can pinpoint which Chinese kilns likely produced the ceramic bowls.

Which choice best states the main idea of the text?

A. Because of a new technology, researchers can locate and recover more shipwrecks than they could in the past.

B. Researchers have been able to identify the location of a number of Chinese kilns in operation 1,000 years ago.

C. With the help of a special tool, researchers have determined the likely origin of bowls recovered from a shipwreck.

D. Before the invention of portable X-ray fluorescence, researchers needed to take a small piece out of an artifact to analyze its components.

ID: cbecb873

Body Length, Filter Time, and Lunges per Dive for Four Whale Species

Whale species	Typical adult body length (meters)	Average time to filter all engulfed water (seconds)	Average number of lunges per dive deeper than 50 meters
fin	18–22	31.30	3.95
humpback	11–17	17.12	6.28
minke	7–10	8.88	7.48
blue	24–34	60.27	4.02

Some whale species practice lunge feeding, in which they lunge toward prey with their mouths open at wide angles, collect the prey and the surrounding water, and then filter out the water through baleen plates in their mouths. Although the volume of water engulfed increases with whales' body length, the surface area of whales' baleen plates, which influences the rate at which water can be filtered, does not increase with body length to the same degree, which helps explain why _____

Which choice most effectively uses data from the table to complete the statement?

A. minke whales and humpback whales show similar average filter times.

B. humpback whales show an average of 6.28 lunges per dive.

C. fin whales show a longer average filter time than minke whales do.

D. blue whales show the longest average filter time and the highest average number of lunges per dive.

ID: 87023f34

"Ghosts of the Old Year" is an early 1900s poem by James Weldon Johnson. In the poem, the speaker describes experiencing an ongoing cycle of anticipation followed by regretful reflection: _____

Which quotation from "Ghosts of the Old Year" most effectively illustrates the claim?

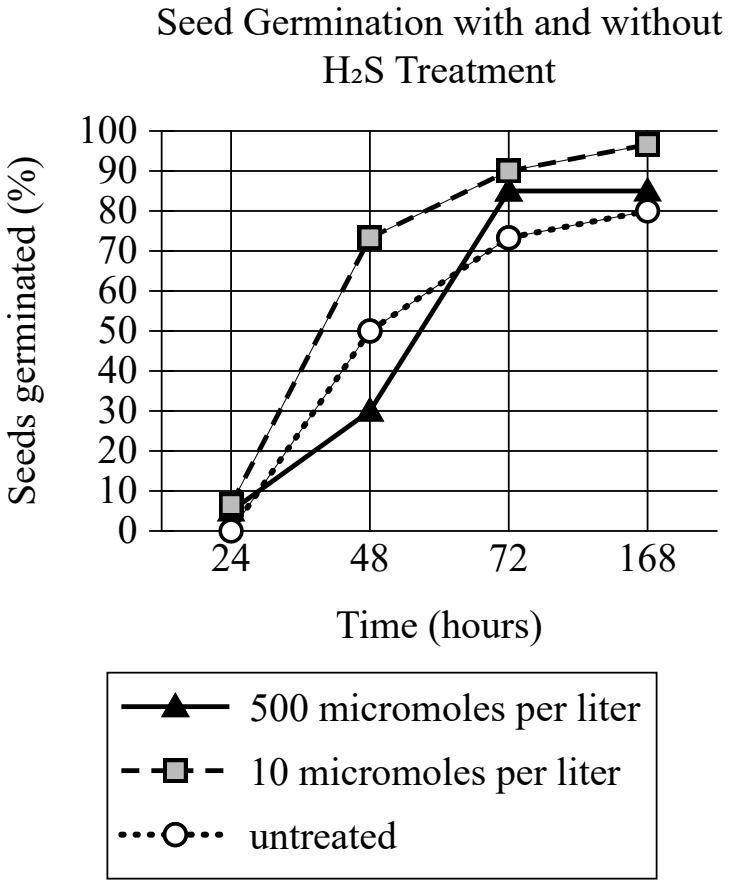
A. "What does this brazen tongue declare, / That falling on the midnight air / Brings to my heart a sense of care / Akin to fright?"

B. "The snow has ceased its fluttering flight, / The wind sunk to a whisper light, / An ominous stillness fills the night, / A pause—a hush."

C. "It tells of many a squandered day, / Of slighted gems and treasured clay, / Of precious stores not laid away, / Of fields unreaped."

D. "And so the years go swiftly by, / Each, coming, brings ambitions high, / And each, departing, leaves a sigh / Linked to the past."

ID: 8a584241



In high concentrations, hydrogen sulfide (H_2S) is typically toxic to many plants. Frederick D. Dooley and colleagues wanted to understand what effects low doses of H_2S might have on plant growth. They treated bean, corn, wheat, and pea seeds with various concentrations (measured in micromoles per liter) of H_2S and tracked the germination of those seeds along with the germination of untreated seeds. Treatment with particular concentrations of H_2S was associated with accelerated germination: for example, _____

Which choice most effectively uses data from the graph to complete the statement?

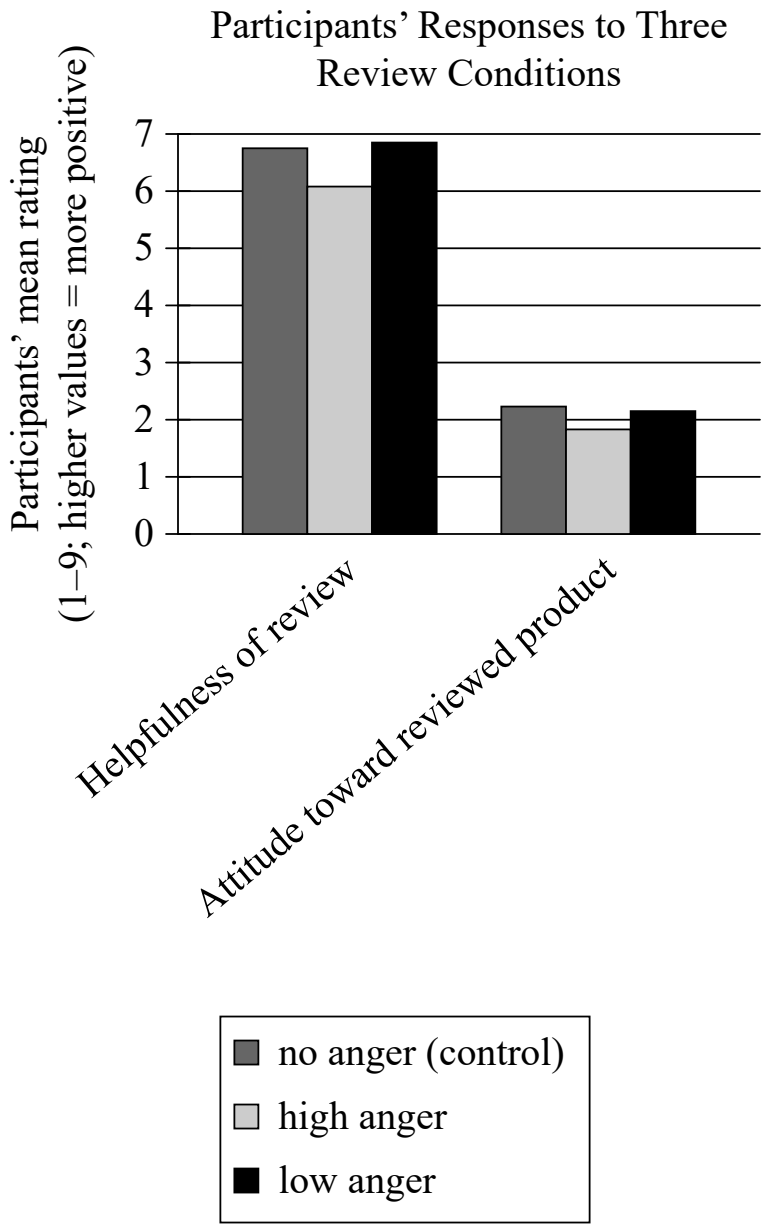
A. at 24 hours, less than 10% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas more than 90% of those seeds had germinated at 168 hours.

B. at 48 hours, more than 70% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 50% of untreated seeds had germinated.

C. at 168 hours, more than 90% of seeds treated with H_2S at concentrations of 10 or 500 micromoles per liter had germinated, whereas less than 70% of untreated seeds had germinated.

D. at 48 hours, approximately 50% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 30% of untreated seeds had germinated.

ID: d5b9ed0d



To understand how expressions of anger in reviews of products affect readers of those reviews, business scholar Dezhi Yin and colleagues measured study participants' responses to three versions of the same negative review—a control review expressing no anger, a review expressing a high degree of anger, and a review expressing a low degree of anger. Reviewing the data, a student concludes that the mere presence of anger in a review may not negatively affect readers' perceptions of the review, but a high degree of anger in a review does worsen readers' perceptions of the review.

Which choice best describes data from the graph that support the students' conclusion?

A. On average, participants' ratings of the helpfulness of the review were substantially higher than were participants' ratings of the reviewed product regardless of which type of review participants had seen.

B. Compared with participants who saw the control review, participants who saw the low-anger review rated the review as slightly more helpful, whereas participants who saw the high-anger review rated the review as less helpful.

C. Participants who saw the low-anger review rated the review as slightly more helpful than participants who saw the control review did, but participants' attitude toward the reviewed product was slightly worse when participants saw the low-anger review than when they saw the no-anger review.

ID: f8befe75

Many intellectual histories of the Black Power movement of the 1960s and 1970s rely heavily on essays and other explicitly ideological works as primary sources, a tendency that can overrepresent the perspectives of a small number of thinkers, most of whom were male. Historian Ashley D. Farmer has shown that expanding the array of primary sources to encompass more types of print material—including political cartoons, advertisements, and artwork—leads to a much better understanding of the movement and the crucial and diverse roles that Black women played in shaping it.

Which choice best describes the main idea of the text?

A. Farmer's methods and research have enriched the historical understanding of the Black Power movement and Black women's contributions to it.

B. Before Farmer's research, historians had largely ignored the intellectual dimensions of the Black Power movement.

C. Other historians of the Black Power movement have criticized Farmer's use of unconventional primary sources.

D. The figures in the Black Power movement whom historians tend to cite would have agreed with Farmer's conclusions about women's roles in the movement.

ID: 1d08c7ee

Pulitzer Prize–winning writer Héctor Tobar has built a multifaceted career as both a journalist and an author of short stories and novels. In an essay about Tobar's work, a student claims that Tobar blends his areas of expertise by applying journalism techniques to his creation of works of fiction.

Which quotation from a literary critic best supports the student's claim?

A. "For one novel, an imagined account of a real person's global travels, Tobar approached his subject like a reporter, interviewing people the man had met along the way and researching the man's own writings."

B. "Tobar got his start as a volunteer for *El Tecolote*, a community newspaper in San Francisco, and wrote for newspapers for years before earning a degree in creative writing and starting to publish works of fiction."

C. "Many of Tobar's notable nonfiction articles are marked by the writer's use of techniques usually associated with fiction, such as complex narrative structures and the incorporation of symbolism."

D. "The protagonist of Tobar's third novel is a man who wants to be a novelist and keeps notes about interesting people he encounters so he can use them when developing characters for his stories."

ID: ed314256

The most recent iteration of the immersive theater experience *Sleep No More*, which premiered in New York City in 2011, transforms its performance space—a five-story warehouse—into a 1930s-era hotel. Audience members, who wander through the labyrinthine venue at their own pace and follow the actors as they play out simultaneous, interweaving narrative loops, confront the impossibility of experiencing the production in its entirety. The play's refusal of narrative coherence thus hinges on the sense of spatial fragmentation that the venue's immense and intricate layout generates.

What does the text most strongly suggest about *Sleep No More*'s use of its performance space?

A. The choice of a New York City venue likely enabled the play's creators to experiment with the use of theatrical space in a way that venues from earlier productions could not.

B. Audience members likely find the experience of the play disappointing because they generally cannot make their way through the entire venue.

C. The production's dependence on a particular performance environment would likely make it difficult to reproduce exactly in a different theatrical space.

D. Audience members who navigate the space according to a recommended itinerary will likely have a better grasp of the play's narrative than audience members who depart from that itinerary.

ID: 4d3e3c52

In a paper about p-i-n planar perovskite solar cells (one of several perovskite cell architectures designed to collect and store solar power), Lyndsey McMillon-Brown et al. describe a method for fabricating the cell's electronic transport layer (ETL) using a spray coating. Conventional ETL fabrication is accomplished using a solution of nanoparticles. The process can result in a loss of up to 80% of the solution, increasing the cost of manufacturing at scale—an issue that may be obviated by spray coating fabrication, which the researchers describe as "highly reproducible, concise, and practical."

What does the text most strongly suggest about conventional ETL fabrication?

A. It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.

B. It is more expensive when manufacturing at scale than are processes for fabricating ETLs used in other perovskite solar cell architectures.

C. It typically entails a greater loss of nanoparticle solution than do other established approaches for ETL fabrication.

D. It is somewhat imprecise and therefore limits the potential effectiveness of p-i-n planar perovskite solar cells at capturing and storing solar power.

ID: 2fdfe002

The following text is adapted from Countee Cullen's 1926 poem "Thoughts in a Zoo."

> They in their cruel traps, and we in ours,
> Survey each other's rage, and pass the hours
> Commiserating each the other's woe,
> To mitigate his own pain's fiery glow.
> Man could but little proffer in exchange
> Save that his cages have a larger range.
> That lion with his lordly, untamed heart
> Has in some man his human counterpart,
> Some lofty soul in dreams and visions wrapped,
> But in the stifling flesh securely trapped.

Based on the text, what challenge do humans sometimes experience?

A. They cannot effectively tame certain wild animals because of a lack of compassion.

B. They cannot focus on setting attainable goals because of a lack of motivation.

C. They quickly become frustrated when faced with difficult tasks because of a lack of self-control.

D. They have aspirations that cannot be fulfilled because of certain limitations.

ID: cef77aa7

Geoglyphs are large-scale designs of lines or shapes created in a natural landscape. The Nazca Lines were created in the Nazca Desert in Peru by several Indigenous civilizations over a period of many centuries. Peruvian archaeologist Johny Isla specializes in these geoglyphs. At a German exhibit about the Nazca Lines, he saw an old photograph of a large geoglyph of a whalelike figure and was surprised that he didn't recognize it. Isla returned to Peru and used a drone to search a wide area, looking for the figure from the air. This approach suggests that Isla thought that if he hadn't already seen it, the whalelike geoglyph _____

Which choice most logically completes the text?

A. must represent a species of whale that went extinct before there were any people in Peru.

B. is actually located in Germany, not Peru, and isn't part of the Nazca Lines at all.

C. is probably in a location Isla hadn't ever come across while on the ground.

D. was almost certainly created a long time after the other Nazca Lines geoglyphs were created.

ID: db2da2bf

In 2019, 20 previously unknown moons were confirmed to be orbiting Saturn. Three of the moons have prograde orbits (orbiting in the direction the planet spins), and the other 17 have retrograde orbits (orbiting in the opposite direction of the planet's spin). All but one of the 20 moons are thought to be remnants of bodies that orbited Saturn until they broke apart in collisions. Although the one exceptional moon orbits in the same direction as the planet's spin, its orbit is highly eccentric compared to the rest, which may suggest that it has a different origin than the other 19 moons.

Based on the text, which choice best describes the moon with the eccentric orbit?

A. It doesn't have a retrograde orbit, but it likely has the same origin as the moons with retrograde orbits.

B. Its orbit is so tilted with respect to the other moons' orbits that it's neither prograde nor retrograde.

C. It has a prograde orbit that is likely the result of having collided with another body orbiting Saturn.

D. It has a prograde orbit and may not be a remnant of an earlier body that orbited Saturn.

ID: 9731a22b

Paleontologist Lucas E. Fiorelli and colleagues have reported the discovery at a mine in Brazil of several egg clutches, partially preserved single eggs, and egg shells from the Late Cretaceous period. The researchers have concluded that the area was once a nesting and breeding site for titanosaurs, a group of sauropod dinosaurs. The finding is significant given the previous lack of known nesting sites in northern regions of South America, which led many paleontologists to assume that titanosaurs migrated south to lay eggs.

What does the text most strongly suggest about the site discovered by the researchers?

A. It is the earliest known example of a titanosaur nesting and breeding site.

B. It was very difficult to excavate given that it was discovered in a mine.

C. It may have been occupied by other sauropods in addition to titanosaurs.

D. It is farther north than any other nesting site discovered in South America.

ID: f1bfbed3

Marta Coll and colleagues' 2010 Mediterranean Sea biodiversity census reported approximately 17,000 species, nearly double the number reported in Carlo Bianchi and Carla Morri's 2000 census—a difference only partly attributable to the description of new invertebrate species in the interim. Another factor is that the morphological variability of microorganisms is poorly understood compared to that of vertebrates, invertebrates, plants, and algae, creating uncertainty about how to evaluate microorganisms as species. Researchers' decisions on such matters therefore can be highly consequential. Indeed, the two censuses reported similar counts of vertebrate, plant, and algal species, suggesting that _____

Which choice most logically completes the text?

A. Coll and colleagues reported a much higher number of species than Bianchi and Morri did largely due to the inclusion of invertebrate species that had not been described at the time of Bianchi and Morri's census.

B. some differences observed in microorganisms may have been treated as variations within species by Bianchi and Morri but treated as indicative of distinct species by Coll and colleagues.

C. Bianchi and Morri may have been less sensitive to the degree of morphological variation displayed within a typical species of microorganism than Coll and colleagues were.

D. the absence of clarity regarding how to differentiate among species of microorganisms may have resulted in Coll and colleagues underestimating the number of microorganism species.

ID: 24c1b7e4

Percentage Point Changes in US Federal Outlays Relative to GDP by Congressional Status

Period	Congressional status	Change in total outlays	Change in nondefense outlays	Change in defense outlays
1981–1988	divided	−0.4	−1.3	0.9
1975–1976	divided	2.7	3.0	−0.3
1977–1980	undivided	0.3	0.6	−0.3
1964–1968	undivided	1.9	1.4	0.5
1969–1974	divided	−1.8	2.1	−3.9

Economist Steve H. Hanke has shown that divided US Congresses—which occur when one party holds the majority in the House of Representatives and another holds the majority in the Senate—tend to accompany reductions in total federal outlays (spending) relative to gross domestic product (GDP), which Hanke interprets to reflect decreases in government size. Hanke calculated the percentage point change in total outlays (encompassing nondefense and defense outlays) for consecutive US Congresses. Hanke has pointed to his calculations as evidence that a divided Congress may be a "necessary but not sufficient condition" for a decrease in government size to occur.

Which choice best describes data from the table that support the underlined claim?

A. The periods of undivided Congresses were associated with increases in nondefense outlays, whereas all the periods of divided Congresses except one were associated with reductions in defense outlays.

B. All the periods of divided Congresses were associated with reductions in total outlays, although two periods were also associated with increases in nondefense outlays.

C. The periods of undivided Congresses were associated with increases in total outlays, whereas all the periods of divided Congresses were associated with reductions in either nondefense outlays or defense outlays.

D. All the periods of divided Congresses except one were associated with reductions in total outlays, whereas the periods of undivided Congresses were associated with increases in total outlays.

ID: 14189fbb

Having written the impassioned call to arms “Letter to the Spanish Americans” in 1791, Peruvian intellectual Juan Pablo Viscardo y Guzmán is often considered a forerunner for the independence movements in Latin America. But Viscardo’s role in history would have remained insignificant were it not for Venezuelan revolutionary Francisco de Miranda, who was handed the unpublished letter after Viscardo’s death. Miranda not only helped circulate the letter, but his edits and footnotes to the text position Miranda as a central figure in the text’s creation.

Which choice best states the main idea of the text?

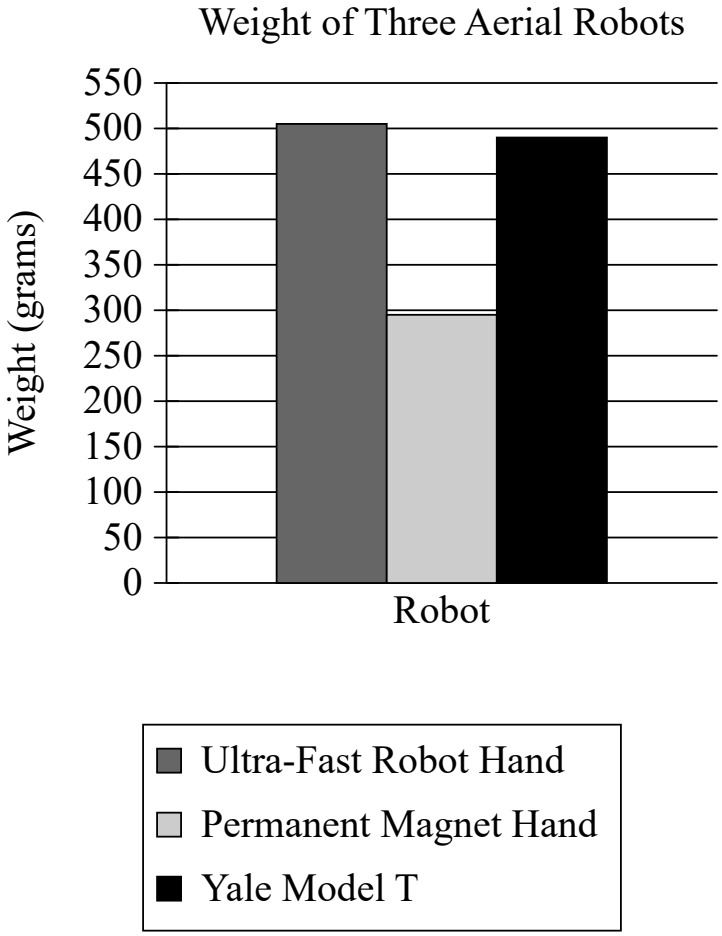
A. The original authorship of “Letter to the Spanish Americans” is disputed by contemporary historians.

B. The majority of the most eloquently stated arguments in “Letter to the Spanish Americans” were written by Miranda.

C. Miranda played a crucial role in influencing the content and distribution of “Letter to the Spanish Americans.”

D. “Letter to the Spanish Americans” persuaded many people in Latin America to pursue national independence.

ID: df37c087



Aerial robots vary considerably in their holding force; the Ultra-Fast Robot Hand, for example, has a holding force of 56 newtons, more than twice that of the Permanent Magnet Hand and more than four times that of the Yale Model T. Since an aerial robot must lift its own weight along with its cargo, engineer Jiawei Meng and colleagues used a ratio of each robot's holding force to the robot's weight to calculate payload capacity, with higher ratios corresponding to greater capacity, concluding that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Which choice best describes data in the graph that support Meng and colleagues' conclusion?

A. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than 450 grams.

B. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than the Permanent Magnet Hand does.

C. The Yale Model T has a lower holding force than the Permanent Magnet Hand despite weighing more.

D. The Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T does.

ID: 4889580c

Archaeologists and historians used to believe that the Maya civilization during its Classic period (roughly 250–900) lacked agricultural marketplaces. One reason for this belief was that these scholars misunderstood the ecology of the regions the Maya inhabited. Marketplaces typically emerge because different individuals or groups want to trade resources they control for resources they don't control. Scholars seriously underestimated the ecological diversity of the Maya landscape and thus assumed that _____

Which choice most logically completes the text?

A. marketplaces likely would not have attracted many traders from outside the regions controlled by the Maya.

B. farming practices would have been largely the same throughout Maya lands even if the crops people produced varied significantly.

C. marketplaces would not have enabled Maya people to acquire many products different from those they already produced.

D. farmers would trade agricultural products only if they had already produced enough to meet their own needs.

ID: 9abc3ba5

“Gestures” in painting are typically thought of as bold, expressive brushstrokes. In the 1970s, American painter Jack Whitten built a 12-foot (3.7-meter) tool he named the “developer” to apply paint to an entire canvas in one motion, resulting in his series of “slab” paintings from that decade. Whitten described this process as making an entire painting in “one gesture,” signaling a clear departure from the prevalence of gestures in his work from the 1960s. Some art historians claim this shift represents “removing gesture” from the process. Therefore, regardless of whether using the developer constitutes a gesture, both Whitten and these art historians likely agree that _____

Which choice most logically completes the text?

A. any tool that a painter uses to create an artwork is capable of creating gestures.

B. Whitten’s work from the 1960s exhibits many more gestures than his work from the 1970s does.

C. Whitten became less interested in exploring the role of gesture in his work as his career progressed.

D. Whitten’s work from the 1960s is much more realistic than his work from the 1970s is.

ID: 39e440e4

Archaeologists have held that the Casarabe culture, which emerged in the southwestern Amazon basin in the first millennium CE, was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness. Recently, however, archaeologist Heiko Prümers and colleagues conducted a study of the region using remote-sensing technology that enabled them to create three-dimensional images of the jungle-covered landscape from above, and the researchers concluded that the Casarabe people developed a form of urbanism in the Amazon basin.

Which finding about the remote-sensing images, if true, would most directly support Prümers and colleagues' conclusion?

A. They show shapes consistent with widely separated settlements of roughly equal small size surrounded by uncultivated jungle.

B. They show shapes consistent with long-distance footpaths running from Casarabe territories to large cities outside the region inhabited by the Casarabe people.

C. They show shapes consistent with scattered small farms created by clearing jungle areas near sources of fresh water.

D. They show shapes consistent with monumental platforms and dense central settlements linked to smaller settlements by a system of canals and roadways.

ID: 3bfcb73b

An archaeological team led by Piotr Bieliński and Sultan al-Bakri found remnants of a 4,000-year-old Bronze Age board game at a site in Oman. Little is left of the game except a stone board, which is carved with a grid and has places to hold game pieces. Some scholars claim that the game was largely played by traders.

Which finding, if true, would most directly support the scholars' claim?

A. Other examples of the game dating to the same period have been found in the remains of several homes in the region, including in one home that may have belonged to a trader.

B. Similar games have been found in other sites dating to the same period that were connected to the site in Oman via trade routes.

C. The other known examples of the game dating to the same period have been found along routes that seem to have been used primarily by traders at the time.

D. Remnants of other goods have been found at the site in Oman that probably also reached the location through trade.

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In superfluorescence, electrical charges known as dipoles emit light in synchronized bursts so intense that they are visible to the eye. Until recently, this phenomenon has only been observed at extremely cold temperatures because dipoles cannot synchronize at higher temperatures. But in a study, Melike Biliroglu and colleagues observed superfluorescence at room temperature in thin films made of perovskite and other similarly crystalline materials; the researchers propose that the formation of shock-absorbing quasiparticles called polarons in the material protects dipoles from thermal interference.

Based on the text, how are polarons believed to be involved in the superfluorescence observed in Biliroglu and colleagues' study?

A. Polarons enable superfluorescent bursts to cross from one crystalline material to another.

B. Polarons allow for the dipoles to synchronize despite higher temperatures.

C. Polarons accelerate the dipoles' release of superfluorescent bursts.

D. Polarons decrease the intensity of the superfluorescent burst.